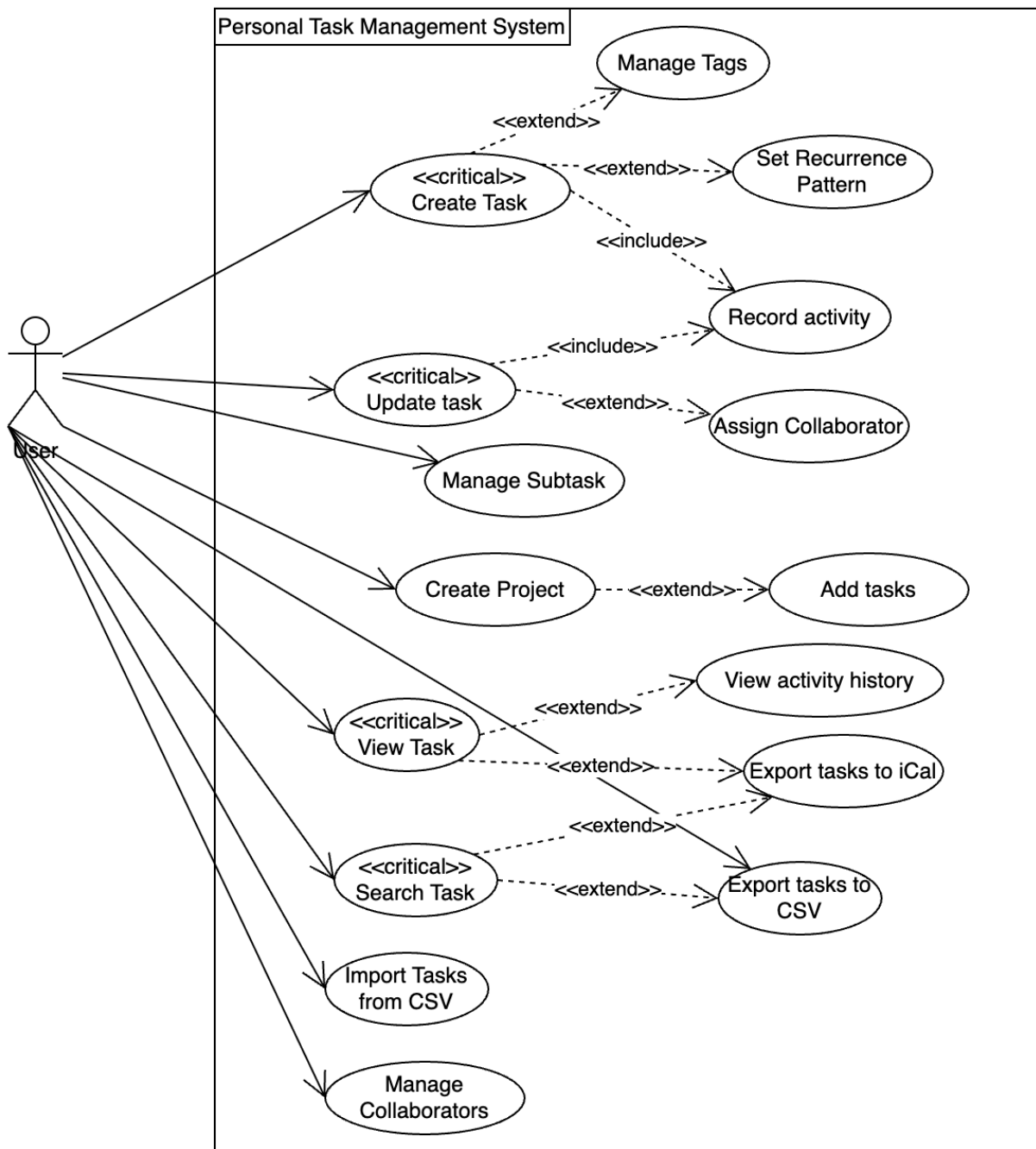
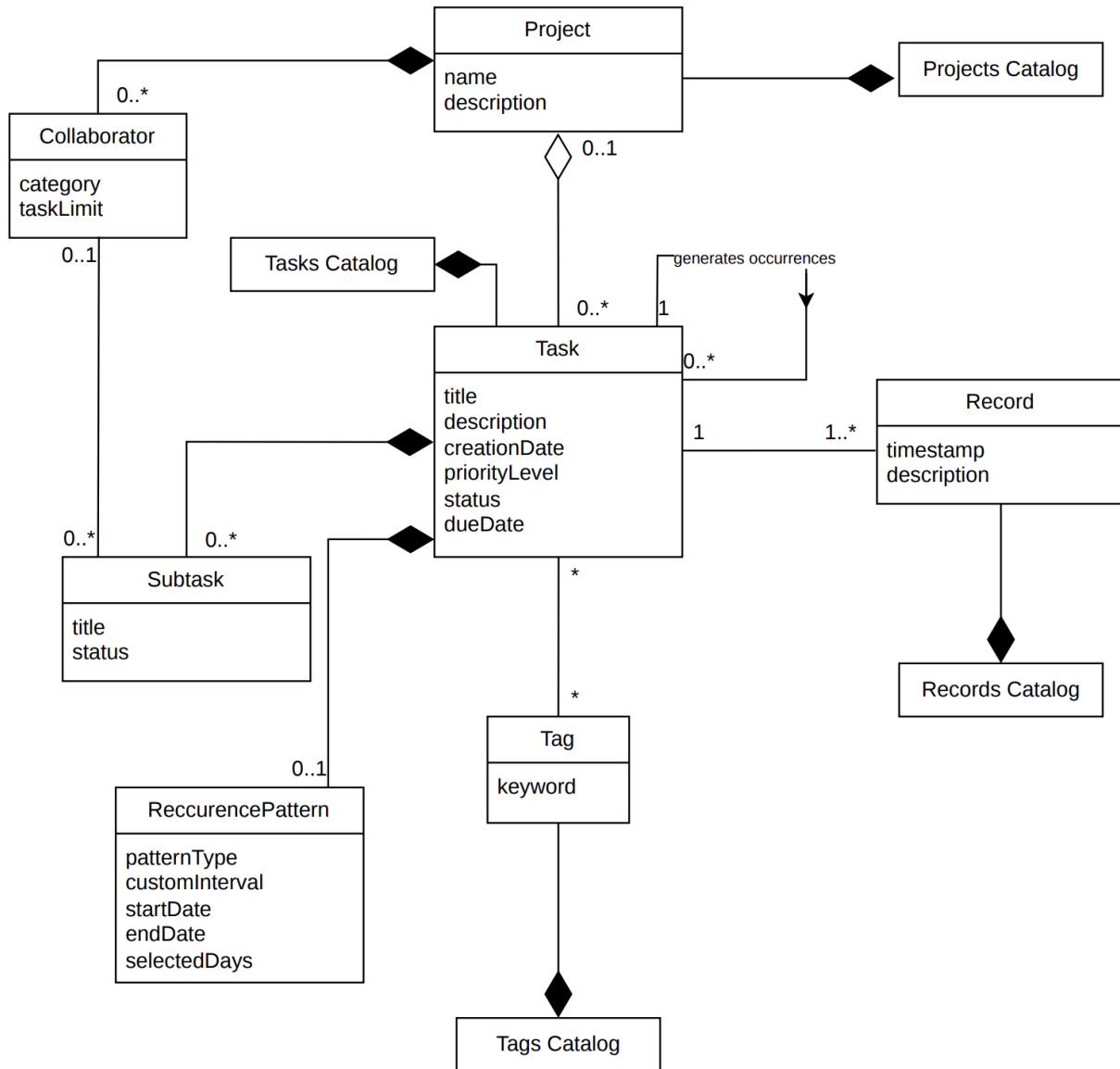


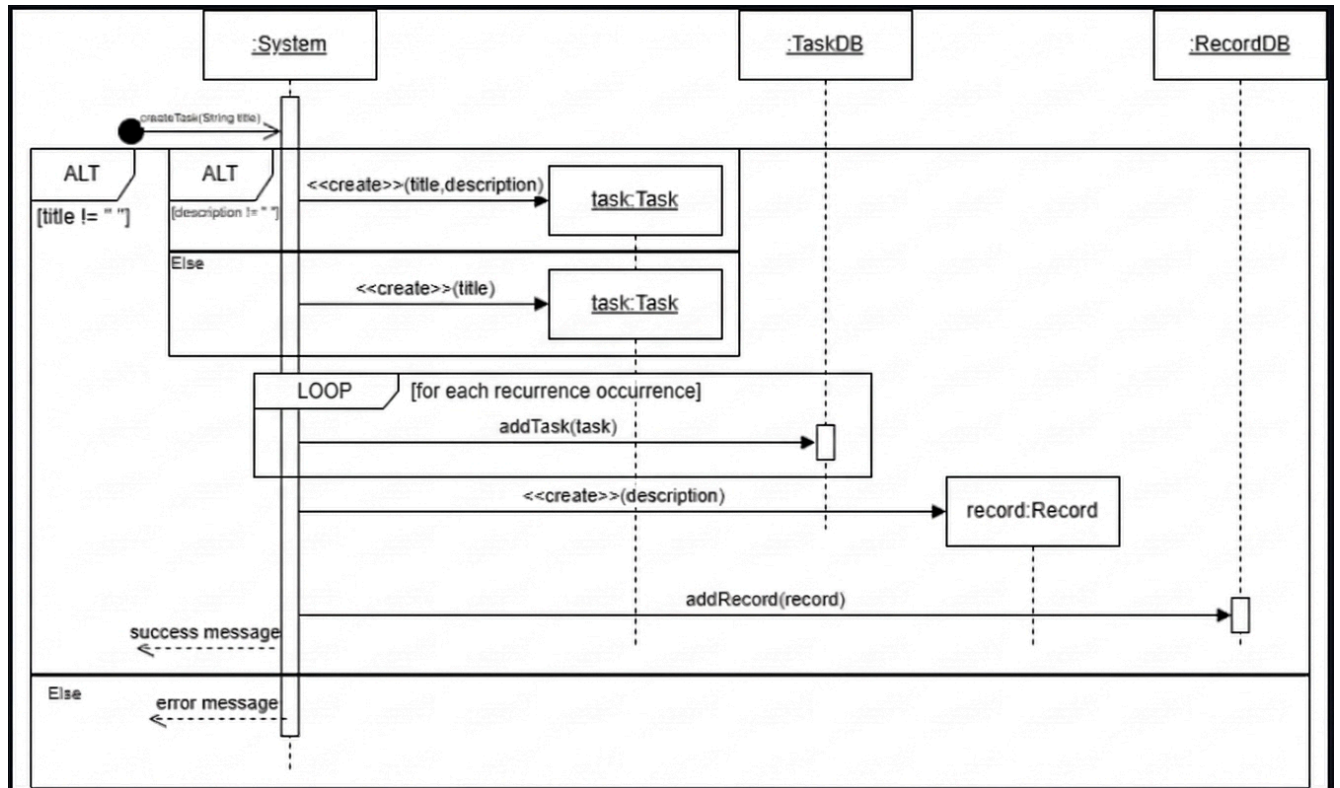
Use Case Diagram



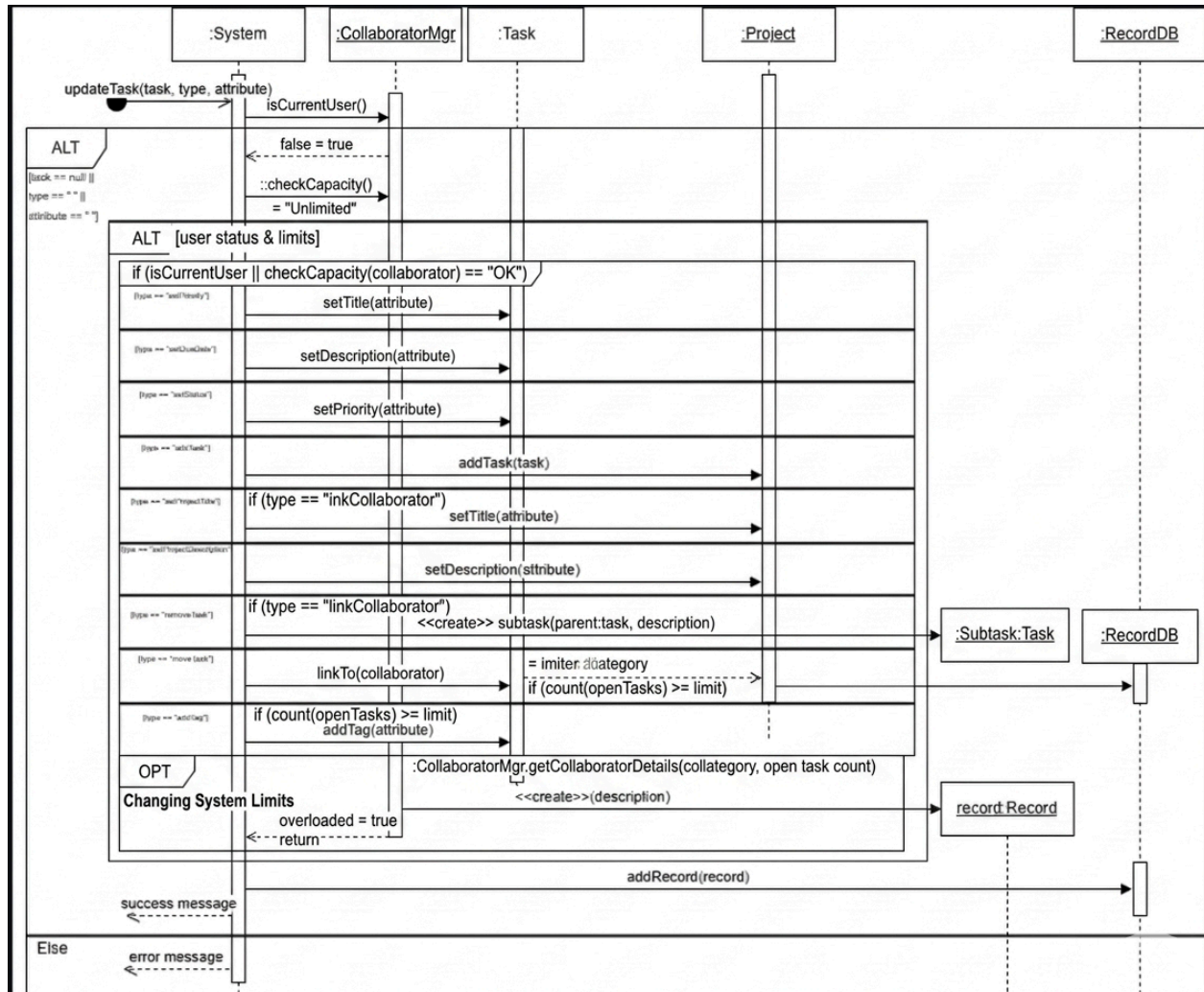
Domain Model



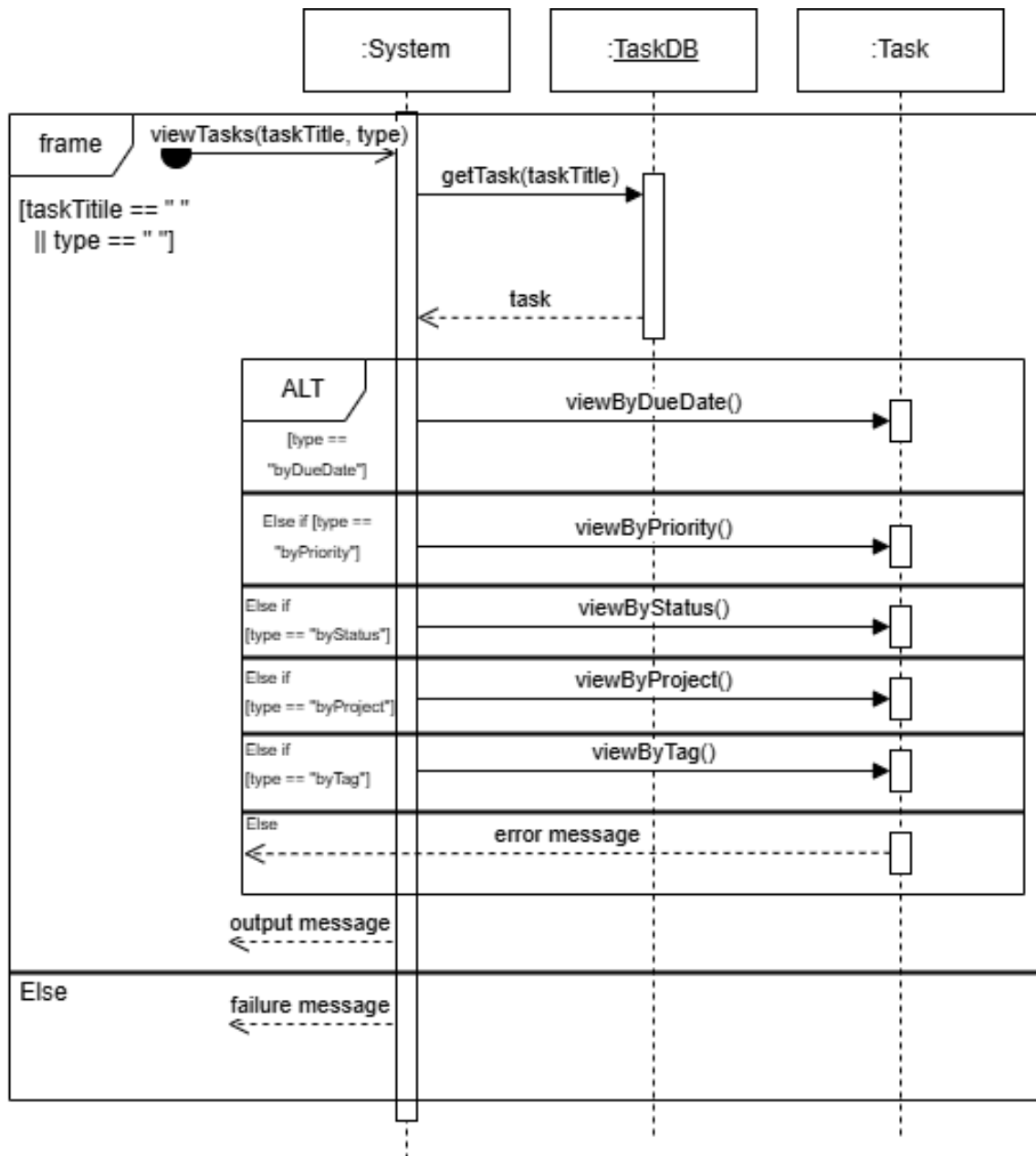
SSD: Create Task



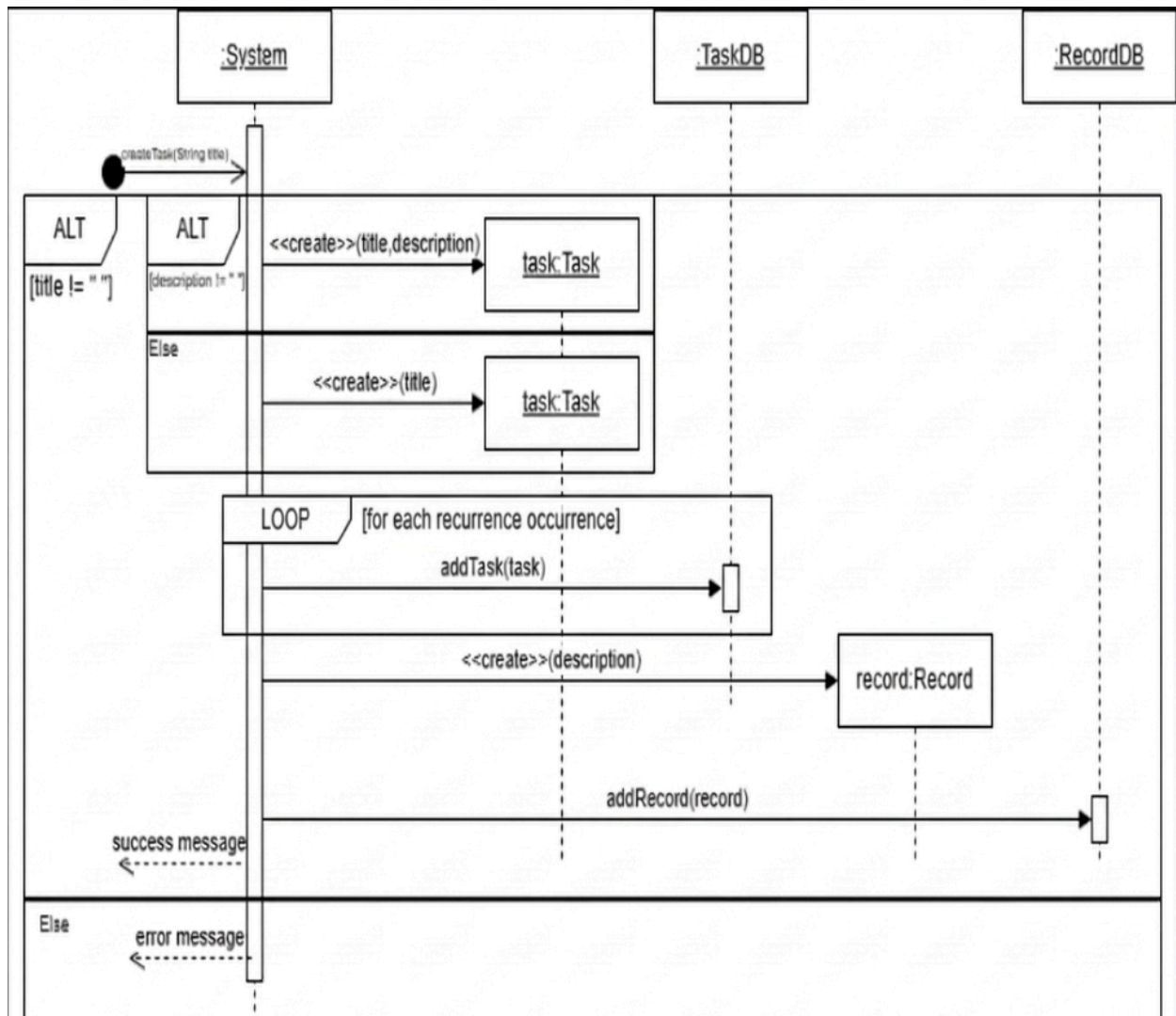
SSD: Update Task



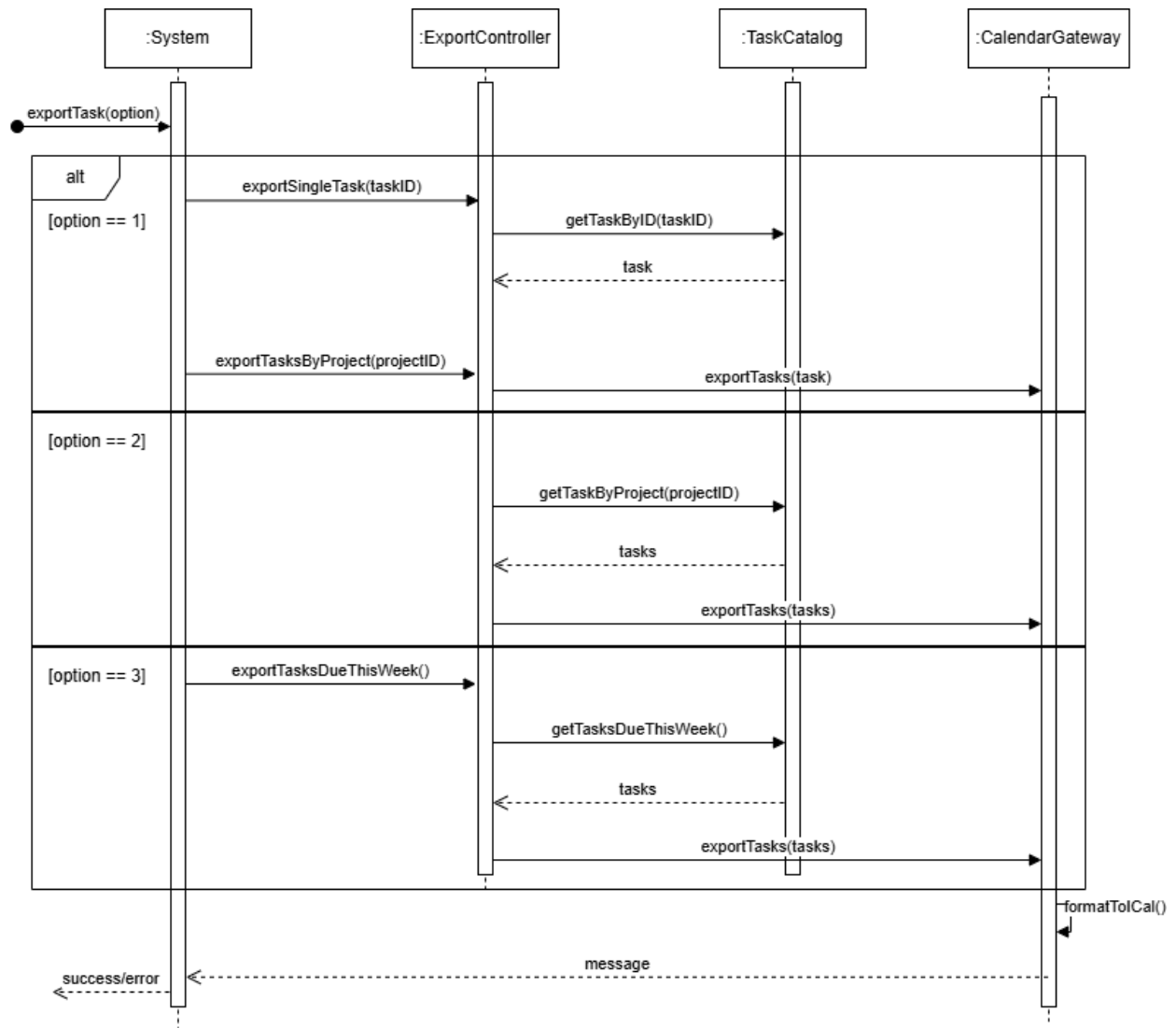
SSD: View Task



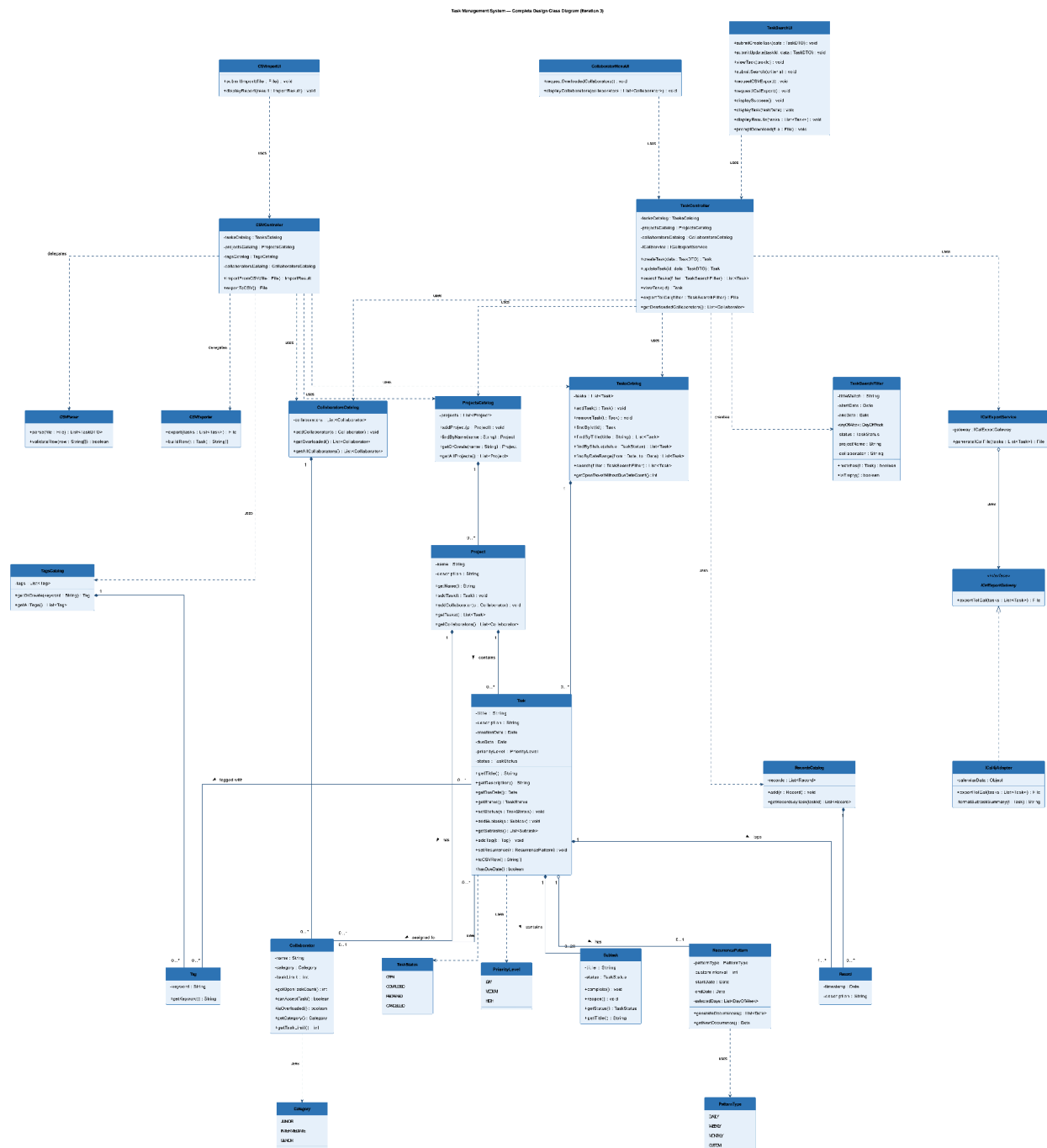
SSD: Search Task



SSD: Export Task to iCal

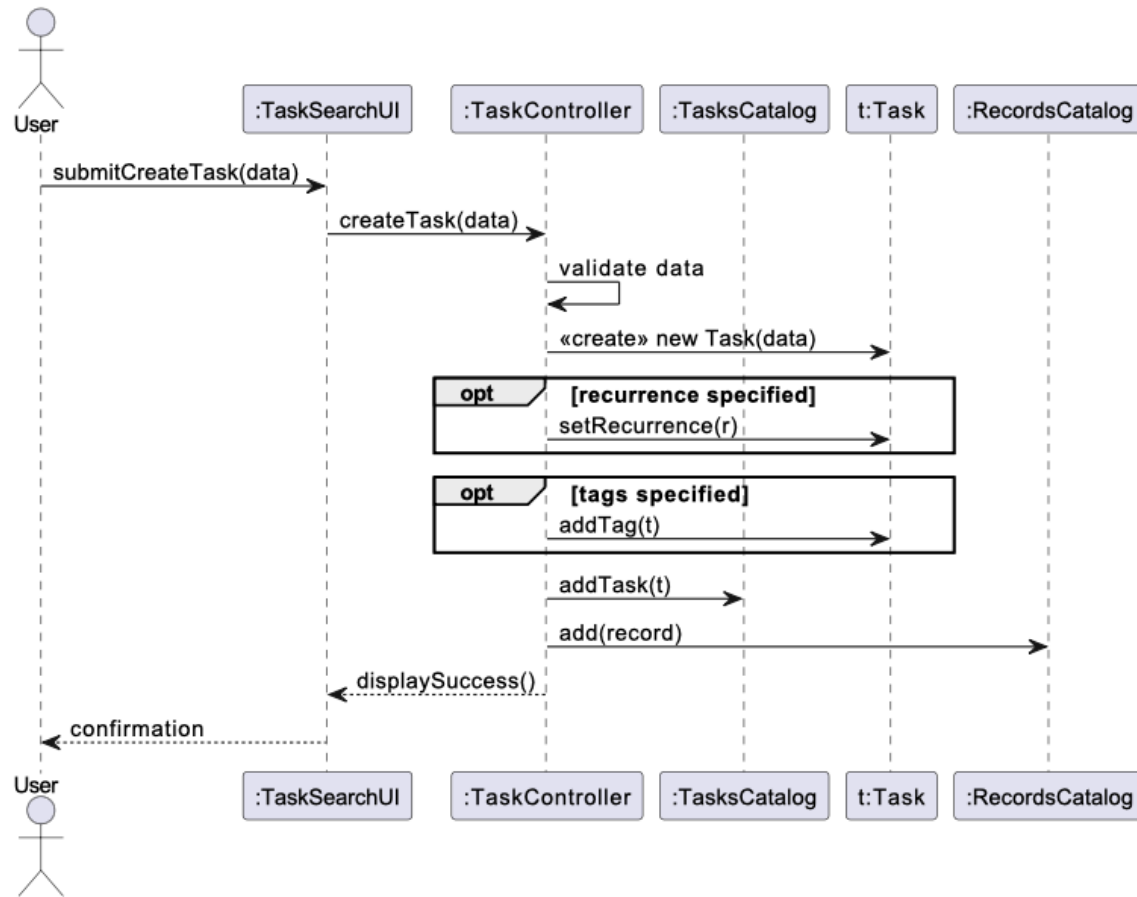


[Link to the SVG/draw.io of the class diagram](#)

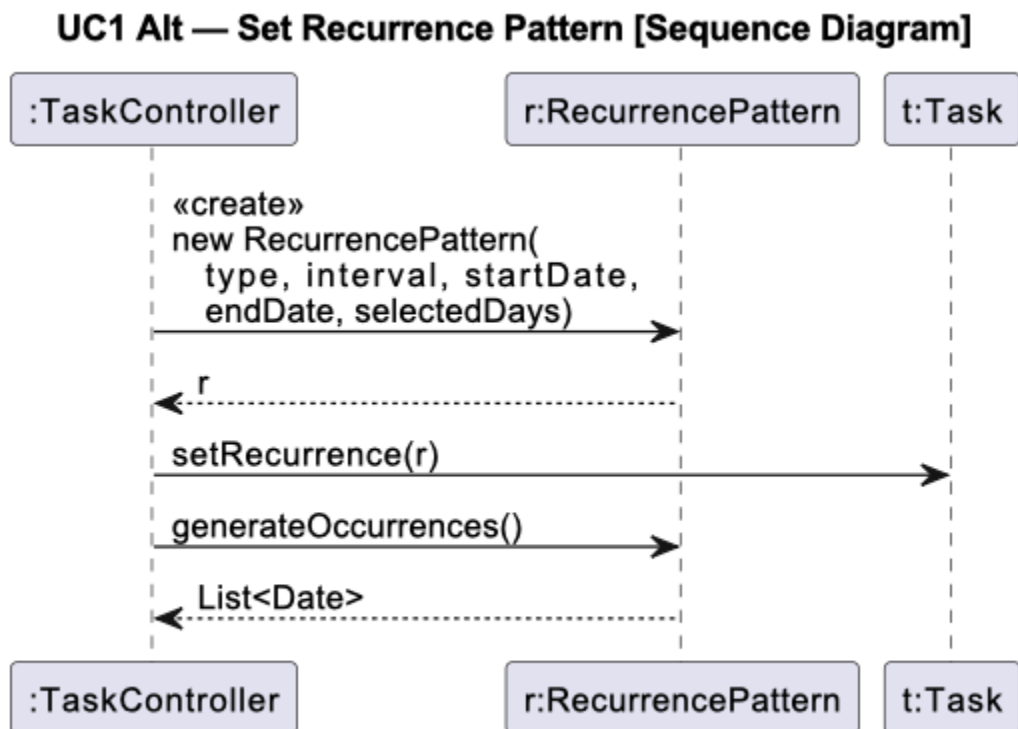


Interaction Diagram: Create Task

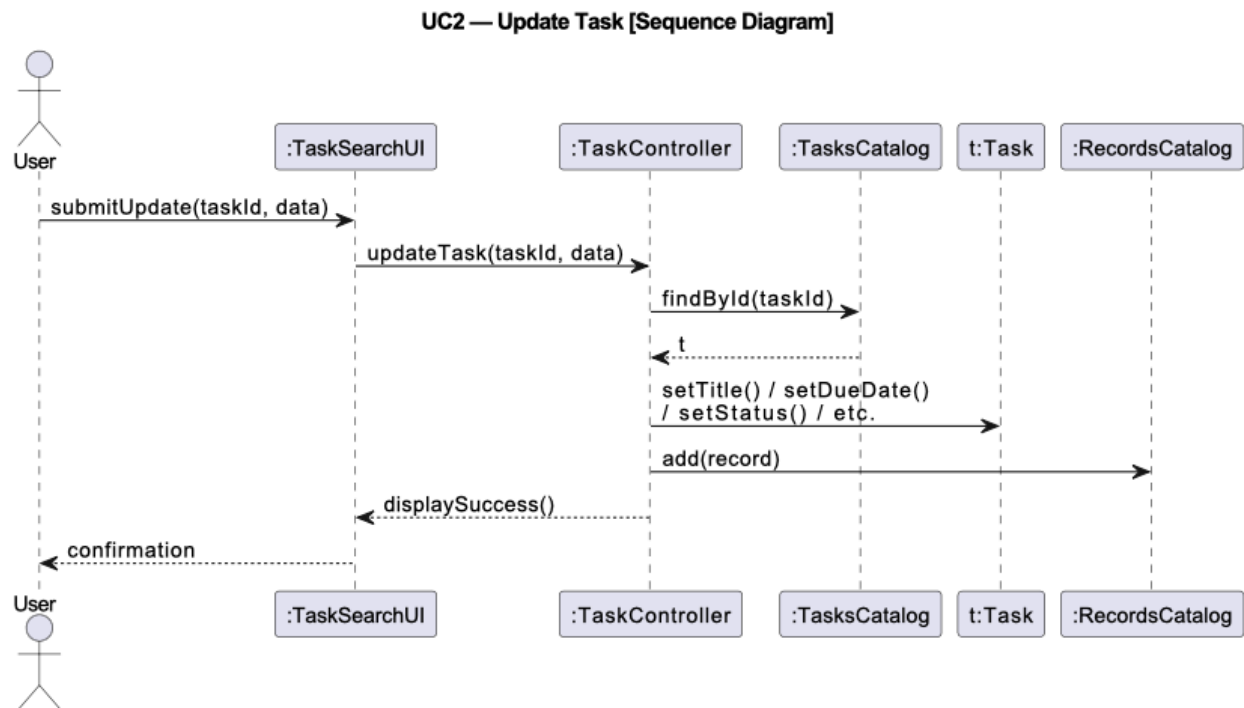
UC1 — Create Task [Sequence Diagram]



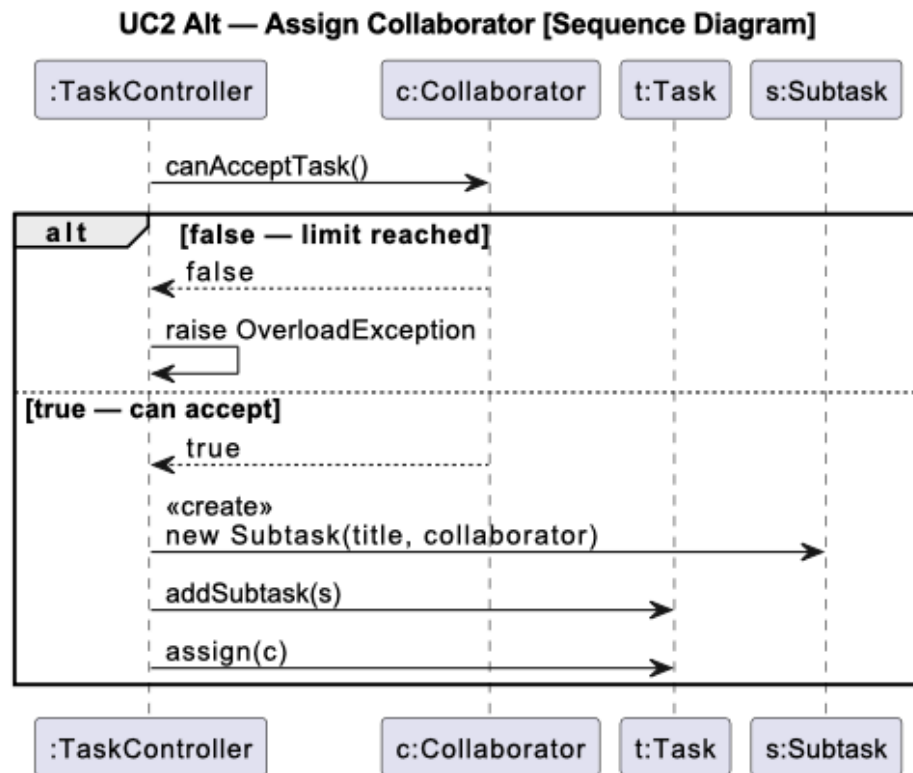
Interaction Diagram: Set Recurrence



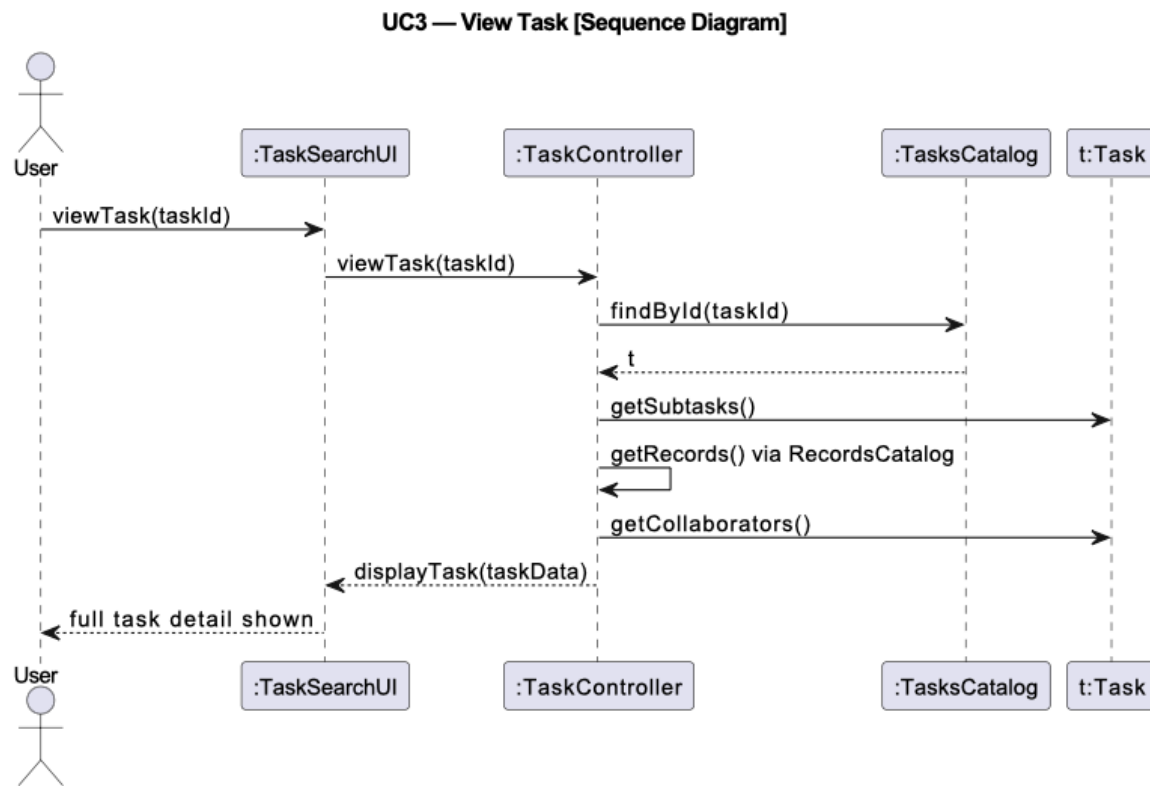
Interaction Diagram: Update Task



Interaction Diagram: Assign Collaborator

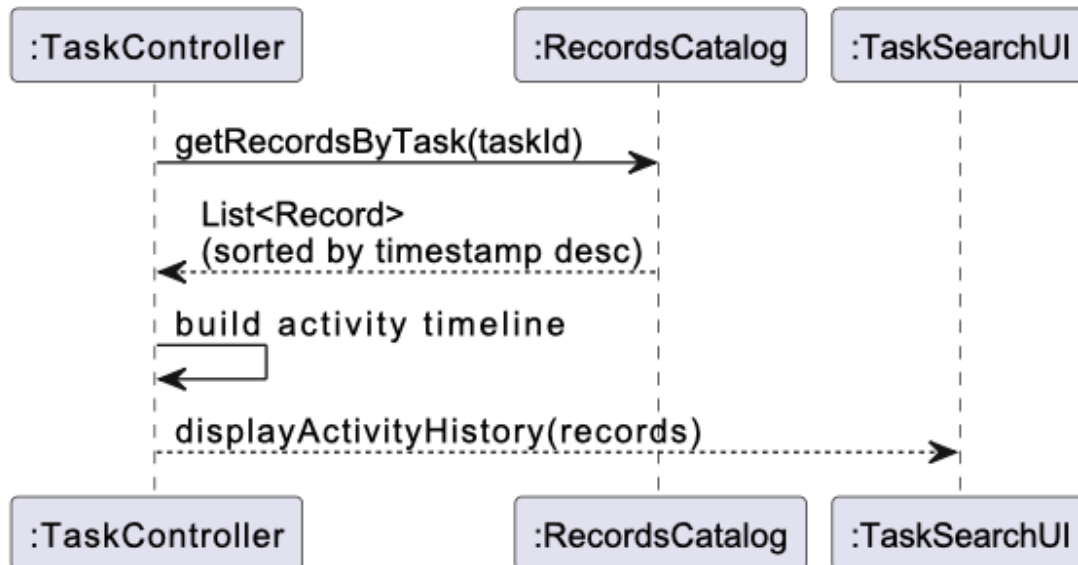


Interaction Diagram: View Task

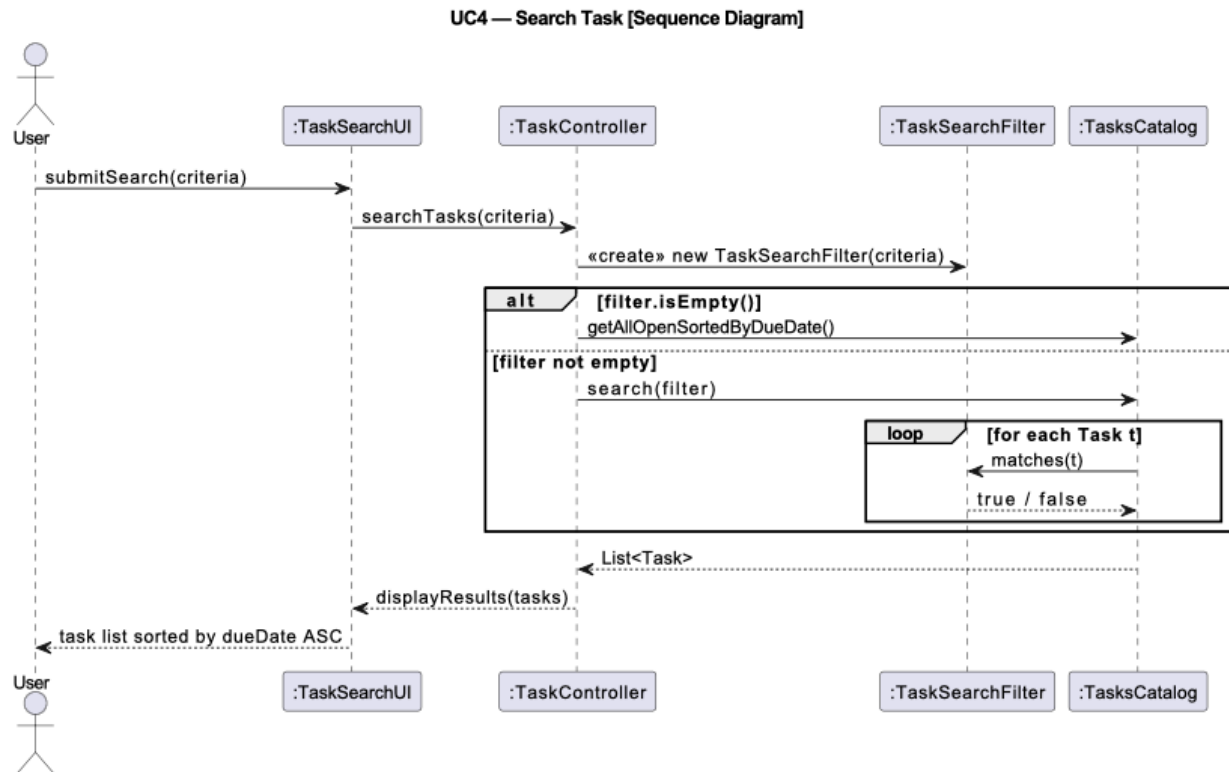


Interaction Diagram: View activity history

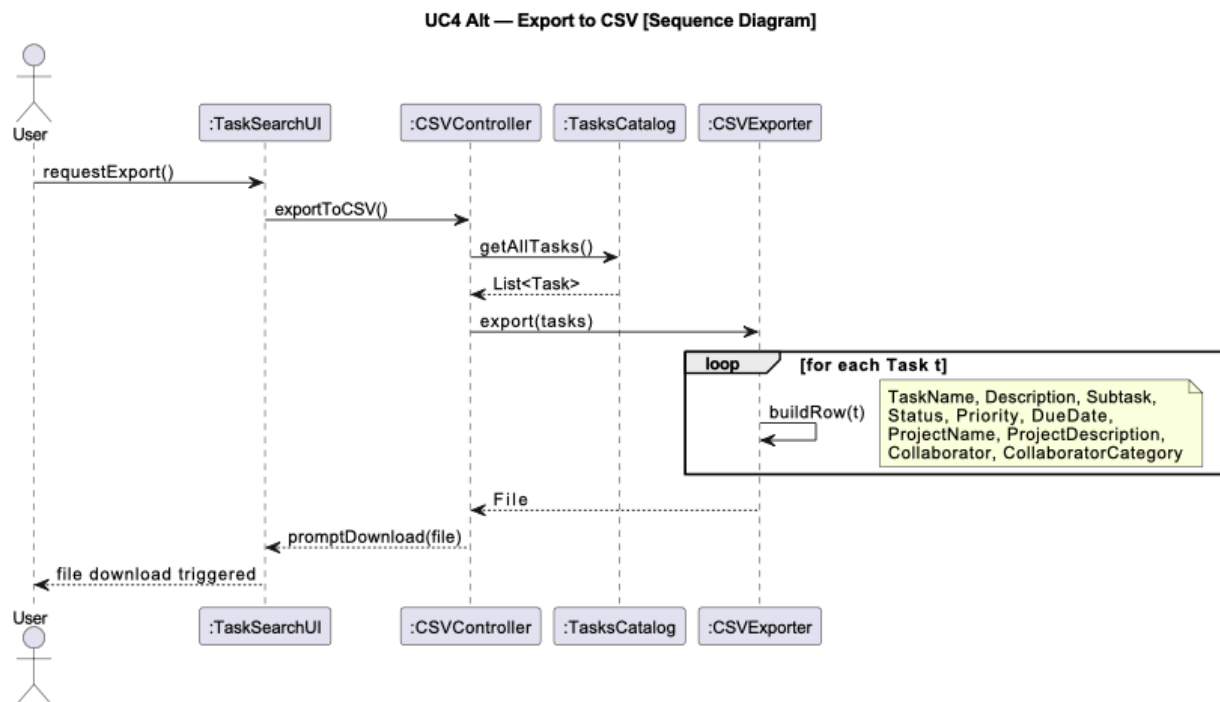
UC3 Alt — View Activity History [Sequence Diagram]



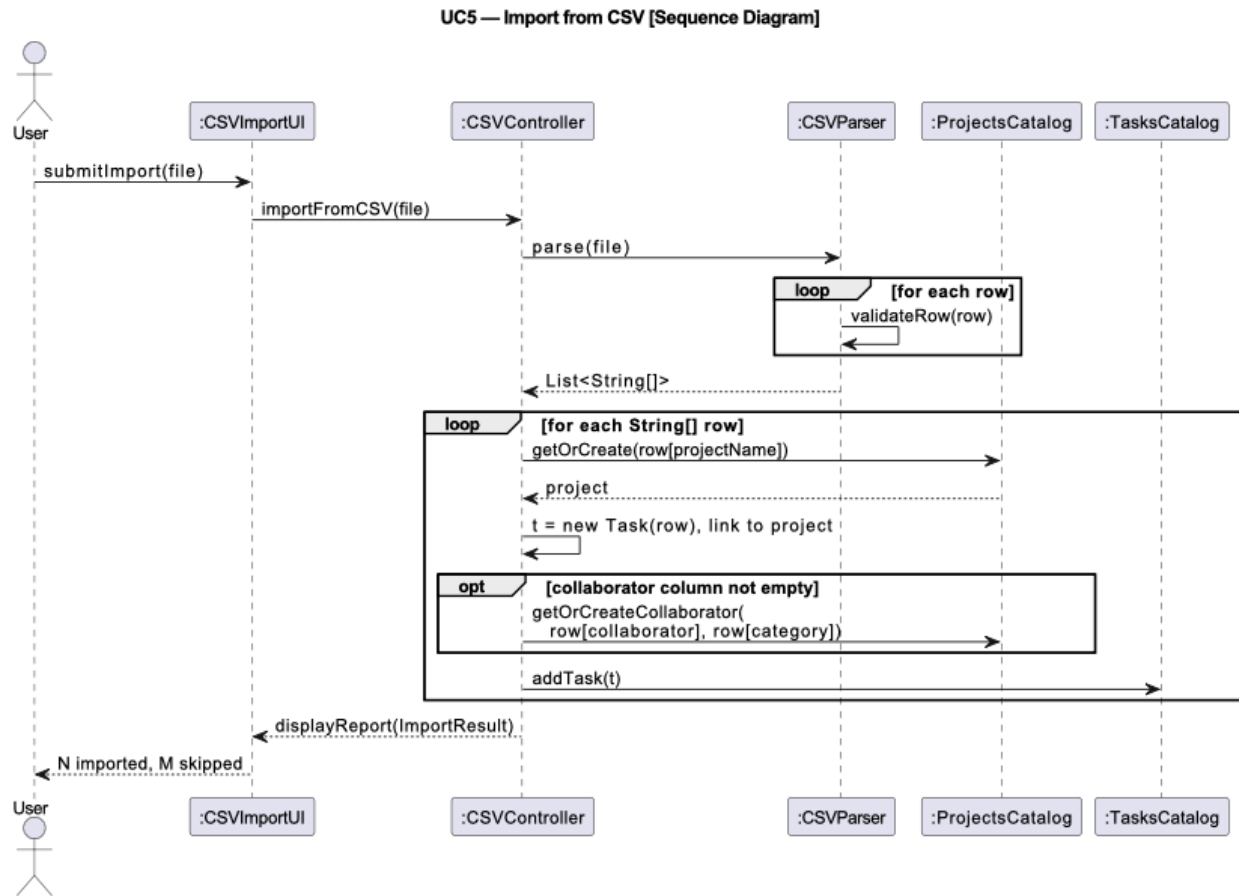
Interaction Diagram: Search Task



Interaction Diagram: Export to CSV



Interaction Diagram: Import from CSV



Other Constraints (OCL)

Constraint 1 — A task cannot have more than 20 sub-tasks

```
context Task
inv: self.subtasks->size() <= 20
```



Constraint 2 — The number of open tasks without a due date should not exceed 50

```
context TasksCatalog
inv: self.tasks
  ->select(t : Task |
    (t.status = TaskStatus::TODO or t.status = TaskStatus::IN_PROGRESS)
    and t.dueDate.oclIsUndefined())
  ->size() <= 50
```



Constraint 3 — The limit for open tasks for each collaborator category is a positive integer

```
context Collaborator
inv: self.taskLimit > 0
```



Constraint 4 — No collaborator must be overloaded

```
context Collaborator
inv: self.tasks
  ->select(t : Task |
    t.status = TaskStatus::TODO or t.status = TaskStatus::IN_PROGRESS)
  ->size() <= self.taskLimit
```



UML [Protocol] State Machine

